

Course Syllabus

General Information

Faculty	Faculty of Arts and Educational Sciences		
Department	Department of Scientific Basic Sciences		
Semester	Second	Academic Year	2024/2025

Course Details

Course Title	Probability and Statistics			Course No.		0182102	
Type of Learning	☐ On- campus ☐ Blended (.....Platform) ☐ Online (.....Platform)						
Credit Hours	3	Theoretical	3	Practical	0		
Course Level (according to JNQF)			7	JNQF Hours*		12	
Pre-requisite	-			Co-requisite		-	

Course Instructor Information

Instructor Name	Fuad A.Abu shaheen,		
Instructor E-mail	fshaheen@meu.edu.jo		
Course Time	(8-9, 9-10) Sunday, Tuesday, Thursday		
Office Hours	(10-12) Sunday, Tuesday, Thursday (9:30-11) Monday, Wednesday		
Office Number	B362	Office Phone	363
Lab Instructor Name –If assigned-			

Sources and References

1. Textbook

Elementary Statistics: A Step-by-Step Approach, Allan G. Bluman, 9th edition.

2. References

Introductory Statistics, Prem S. Mann, 5th edition.
https://archive.org/details/elementarystatis0000blum_x3s4

Teaching Methods

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|---------------------------|--|
| 1- Lectures | 2- Discussion and problem solving |
| 3- Individual assignments | 4- Any active learning method such as: small group |

Course Description

Statistics: descriptive and inferential statistics; **Statistical measures:** measures of central tendency, measures of variation, measures of position; Chebyshev's theorem and empirical rule; **Probability:** definition of probability, laws of probability, Conditional probability, Independence, Counting methods, Combinations and permutations, **Random variables:** definition and types, Expectation; **Distributions:** Binomial distribution, Poisson distribution, Normal distribution; **Sampling distributions:** Confidence intervals, Hypothesis testing.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Competency
1	Calculate statistical measures. CLO Coverage: 40%	PLO 1	K1-P		
2	Calculate theoretical, empirical probabilities and counting rules. CLO Coverage: 25%	PLO 1	K1-P		
3	Calculate the mean, variance, standard deviation, and probabilities for valid probability distributions. CLO Coverage: 20%	PLO 1	K1-P		
4	Build Statistical inference to determine confidence interval and hypothesis testing related to the mean. CLO Coverage: 15%	PLO 1		S1-D	

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative/ Formative
1	Introduction Descriptive and Inferential Statistics	Lecture (1)	On-campus	1	Quiz#1 H.w#1 Midterm Exam Final Exam	Summative
		Lecture (2)	On-campus	1		
2	Variables and Types of Data Organizing Data	Lecture (1)	On-campus	1	Quiz#1 H.w#1 Midterm Exam Final Exam	
		Lecture (2)	On-campus	1		
3	Histograms, Frequency Polygons, and Ogives Measures of Central Tendency	Lecture (1)	On-campus	1	Quiz#1 H.w#1 Midterm Exam Final Exam	
		Lecture (2)	On-campus	2		
4	Measures of Variation	Lecture (1)	On-campus	2	Quiz#1 H.w#1 Midterm Exam Final Exam	
		Lecture (2)	On-campus	2		
5	Measures of Position	Lecture (1)	On-campus	2	Quiz#2 H.w#2 Midterm Exam Final Exam	
		Lecture (2)	On-campus	2		
6	Sample Spaces and Probability the Addition Rules for Probability	Lecture (1)	On-campus	3	Quiz#2 H.w#2 Midterm Exam Final Exam	
		Lecture (2)	On-campus	3		
7	The Multiplication Rules and Conditional Probability	Lecture (1)	On-campus	3	Midterm Exam Final Exam	
		Lecture (2)	On-campus	3		
8	Midterm Exam					
9	Bayes' Theorem Counting Rules	Lecture (1)	On-campus	3	H.w#3 Final Exam	Summative
		Lecture (2)	On-campus	3		

	Counting Rules Probability and Counting Rules	Lecture (1)	On-campus	3	H.w#3	Summative
		Lecture (2)	On-campus	3	Final Exam	
11	Probability Distributions Mean, Variance, Standard Deviation, and Expectation	Lecture (1)	On-campus	2, 3	H.w#3	
		Lecture (2)	On-campus	4	Final Exam	
12	The Binomial Distribution Other Types of Distributions	Lecture (1)	On-campus	4	H.w#3	
		Lecture (2)	On-campus	4	Final Exam	
13	Normal Distributions Applications of the Normal Distribution Distribution of Sample Means	Lecture (1)	On-campus	4	Final Exam	
		Lecture (2)	On-campus	5		
14	Distribution of Sample Means Confidence Intervals for the Mean When s Is Known and unknown	Lecture (1)	On-campus	5	Final Exam	
		Lecture (2)	On-campus	5		
15	Steps in Hypothesis Testing—Traditional Method z Test for a Mean t Test for a Mean	Lecture (1)	On-campus	6	Final Exam	
		Lecture (2)	On-campus	6		
16	Final Exam					

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
Mid-Exam	25%	Week 8
Quizzes	20%	Week 5+ Week 12
Homework Assignments	15%	Week 5+ Week 7+ Week 12
Final Exam	40%	Week 16

Course Policies

Item	Policy
Quizzes	No makeup quizzes. A minimum 3 quizzes are given.
Exams	Midterm and Final exams are given on campus. It is the responsibility of the student to bring all materials needed for the exam such as pens, pencils, calculators, etc. Students are not allowed to borrow anything from each other during the exam.
Makeup Exams	Makeup exam will not be given unless there is a valid excuse.
Drop Date	Last day to drop the course is according to the University regulations.
Academic Honesty	- Cheating or copying on exam or quiz is an illegal and unethical activity. - Standard MEU policy will be applied. All graded assignments must be your own work (your own words).
Attendance	- Excellent attendance is expected. - MEU policy requires the faculty member to assign ZERO grade if a student misses 15% or more of the classes. - Sign-in sheets will be circulated. If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
E-Learning	It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	The student should expect to spend a minimum of 6 hours per week (outside class time) as an average workload.
Graded Exams	Instructor should return exam papers graded to students within one week after the exam date.
Participation	Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any questions. Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	- Homework's are as mentioned in the previous table. - Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. More details about phases and topics of the projects will be given later on.
Others	

Rubrics and Feedbacks (If Applicable)

Key answer

*** JNQF Hours Calculation**

Activity	Duration	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	3	13	39
Tutorials	-	-	-
laboratory	-	-	-
Activities for Assessment and Evaluation			
First Exam/ Midterm	1	1	1
Second Exam (If applicable)	-	-	-
Final Exam	2	1	2
Quizzes	1	3	3
Self-Learning Activities			
Assignments/ Homework	4	2	4
Case Studies	-	-	-
Presentations	-	-	-
E-learning	3	13	39
Extra Readings	2	14	28
Total Hours	120		
JNQF Hours	=Total Hours/10 =12		